



ADITYA RAJ | 25AG65R08

AQUACULTURAL ENGINEERING

EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	M.TECH	IIT Kharagpur	8.37 / 10
2025	B. Tech(Agril. Engg.)	Vasantrao Naik Marathwada Krishi Vidyapeeth, Parbhani	7.93 / 10
2019	Senior Secondary Examination	Bihar School Examination Board, Patna	76.4%
2017	Secondary School Examination	Central Board Of Secondary Education	9.2 / 10

INTERNSHIPS AND PROJECTS

Techno-Economic Optimization of Aqua Feed Formulation & Manufacturing using Alternative Protein Sources	[MTP PROJECT]
- Screened and evaluated alternative protein ingredients for nutritional fit, availability and cost in partial fish meal/soybean meal replacement Optimized feed formulations through graded bench/pilot trials, validating nutritional and pellet quality against existing plant specifications - Conducted techno-economic analysis of formulation, manufacturing, energy costs for cost reduction, payback & implementation strategy.	
AquaPredict: Spatiotemporal Water Quality Prediction & Early Warning System	[Self Project]
- Built an end-to-end ML pipeline integrating water-quality, weather and rainfall data across 5 WB districts for spatiotemporal forecasting - Developed & deployed Streamlit dashboard with LightGBM & XGBoost enable real-time water-quality prediction & automated model training	
Impact of Soil Quality and Weather Parameters on Crop Growth Analysis using Python & Machine Learning	[Self Project]
Analyzed 2.2K+ agricultural records across 7 soil & weather parameters using Python, identifying crop-specific nutrient and climatic patterns. - Executed EDA & correlation analysis with 6+ visualization techniques, deriving nutrient-weather insights for crop suitability & recommendation	
GIS-Based Tourist Map and Route Analysis of Patna City M.Tech Project, IIT Kharagpur Supervisor: Prof. Subhadip Dey	
Developed a GIS-based tourist map of Patna city with route and accessibility analysis for optimal connectivity in major tourist locations	
Stock Price Prediction using LSTM [Long Short-Term Memory] based time series modeling and multi-step forecasting	[Self Project]
Built LSTM time-series forecasting model on 4.7K+ S&P 500 observations, using 50-step sequences of price & volume with an 85:15 train-test split - Engineered 3-layer LSTM with 100 units/layer & 20% dropout implementing multi-step forecasting & evaluating performance using MSE & RMSE	
NEXA[Next-Generation Enhanced Execution Assistant]: Voice-Activated Virtual Assistant Using Python	[Self Project]
- Developed a Python-based voice-activated virtual assistant integrating speech recognition technologies for intelligent user interaction - Enabled voice-controlled automation of routine tasks using speech recognition and command processing for efficient command execution	
Design, Development & Performance Evaluation of Manually Operated Two-Row Trolley-Type Vegetable Transplanter	[Jan'24-May'25]
Designed and fabricated a two-row manual vegetable transplanter, optimizing components for uniform planting, 6 cm depth, and 10 cm hole dia. - Conducted field trials at 0.36 km/h achieving 90.1% effective field efficiency, 0.024 ha/h field capacity with 22 seedlings in 1.41.8 min per 7.5sqm - Reduced transplanting time by 49%, cutting labour from 144 to 2 man-hr/ha & operating cost to 1,710/ha, demonstrating economic feasibility	

SKILLS AND EXPERTISE

Software & Tools : SolidWorks | MATLAB | ArcGIS | QGIS | Jupyter Notebook | VS Code | MS Office | Git | GitHub | Power BI | PostgreSQL
Language & Libraries : C | C++ | Python | NumPy | Pandas | Matplotlib | Seaborn | Scikit-learn | TensorFlow | Keras | Django | LightGBM
Core Skills : Machine Learning | Deep Learning | Data Structures & Algorithms | Statistical Analysis | SQL | EDA | IoT & Embedded

TRAINING

Agricultural Machinery Training, Ministry of Agriculture & Farmers Welfare, Govt. of India; NERFMTTI, Assam	[Aug'23-Sep'23]
Developed hands-on proficiency in practical operation and field application of tractors, power tillers, and an associated farm implements.	
Vocational training on tractor and farm implement operation and maintenance at TAFE CAET, VNMKV, Parbhani	[Aug'24-Sep'24]
Gained practical exposure to construction, working principles, operational procedures, field operation of tractors and farm implements.	
Academic training in AutoCAD drafting and design (2D & basic 3D) at Aditi CAD Training Centre, Parbhani	[Sep'24-Oct'24]
Acquired hands-on skills in 2D drafting, technical drawings, precise dimensioning, layering, and detailed layout preparation using AutoCAD	

COURSEWORK INFORMATION

OPEN CHANNEL HYDRAULICS AND COASTAL ENGG. | FISHERY BIOLOGY & FISH CULTURE TECHNIQUES | FISHERY BIOLOGY AND FISH CULTURE LAB | AQUACULTURAL ENGINEERING LAB. I | SEMINAR | ZERO WASTE FOOD PROCESSING INDUSTRY | WATER QUALITY MANAGEMENT PRACTICE | PRINCIPLES OF FISHING TECHNOLOGY | DESIGN OF AQUACULTURAL FACILITIES AND EQUIPMENT | PLANNING AND DESIGN OF AQUACULTURAL FARMS | AQUACULTURAL ENGINEERING LAB. II | FOOD HANDLING & PACKAGING | PROCESSING AND PRESERVATION OF AQUACULTURAL PRODUCTS | ADVANCED AQUACULTURE TECHNOLOGY

CERTIFICATIONS

Python for Data Science | SQL & Database Management | Machine Learning | Deep Learning & TensorFlow | Data Visualization & Power BI

POSITIONS OF RESPONSIBILITY

Placement Coordinator Aquacultural Engineering Specialization Agricultural and Food Engineering Dept IIT Kharagpur	[Jul'25-Present]
Spearheaded industry-student engagement, managing placement drives, employer interactions, activities and stakeholder coordination	
Teaching Assistant (TA) Agricultural & Food Engineering, IIT Kharagpur	[Sep'26-Present]
Supported faculty in academic delivery, coordinating laboratory sessions, resolving student queries & managing course-related activities	
Hall Secretary (Technology) LBS Hall IIT Kharagpur	[2025-2026]
Spearheaded technical activities & led cross-functional teams at the hall level, driving technology initiatives and coordinating execution	

EXTRA CURRICULAR ACTIVITIES

Smart India Hackathon Volunteer | NSS State-Level Disaster Management Camp | Inter-Hall Sports Representative, IIT Kharagpur